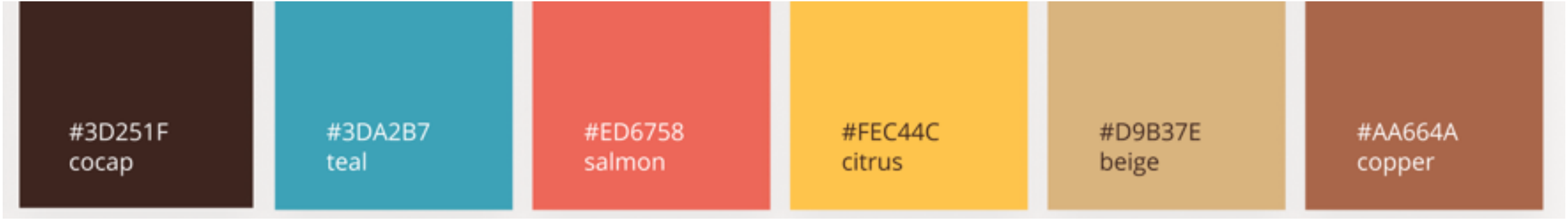


Sinkhorn GAN



#3D251F
cocap

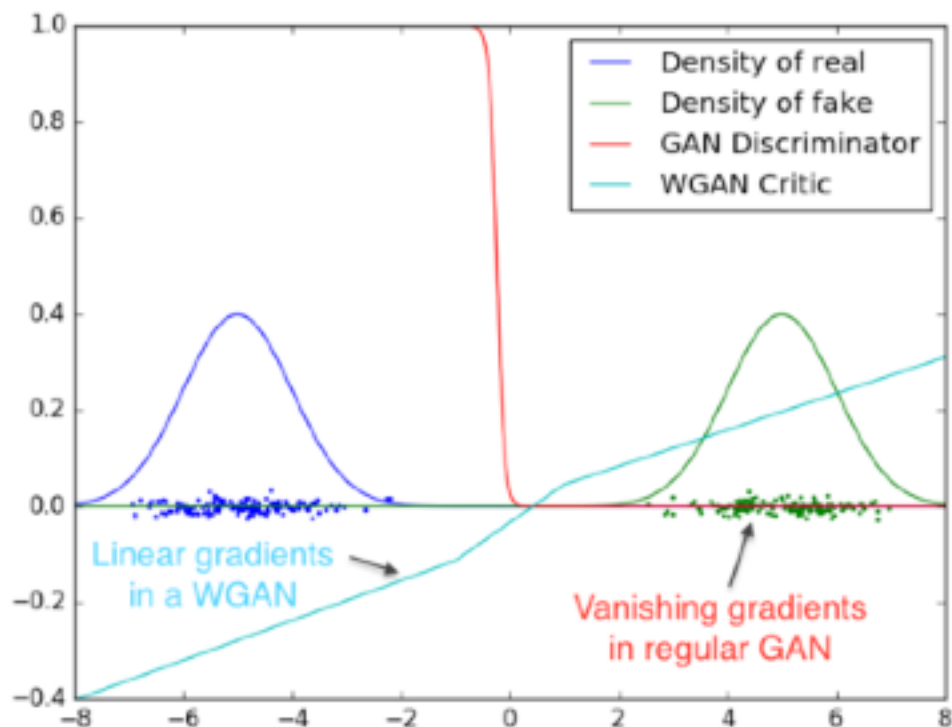
#3DA2B7
teal

#ED6758
salmon

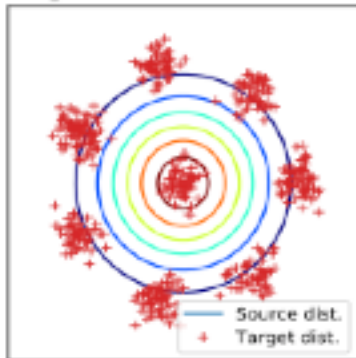
#FEC44C
citrus

#D9B37E
beige

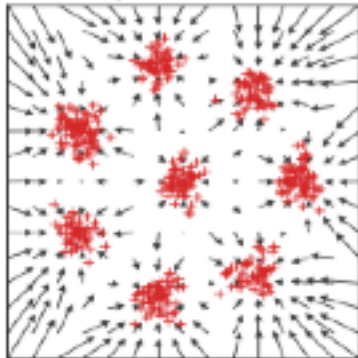
#AA664A
copper



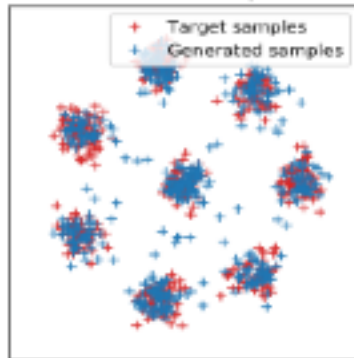
Target and Source distributions



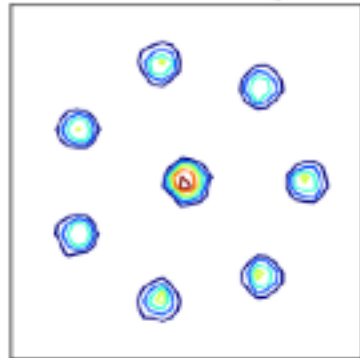
Displacement field



Generated Samples



Generated density



Learning Generative Models with Sinkhorn Divergences

Aude Genevay

CEREMADE,

Université Paris-Dauphine

Gabriel Peyré

CNRS and DMA,

École Normale Supérieure

Marco Cuturi

CREST ENSAE

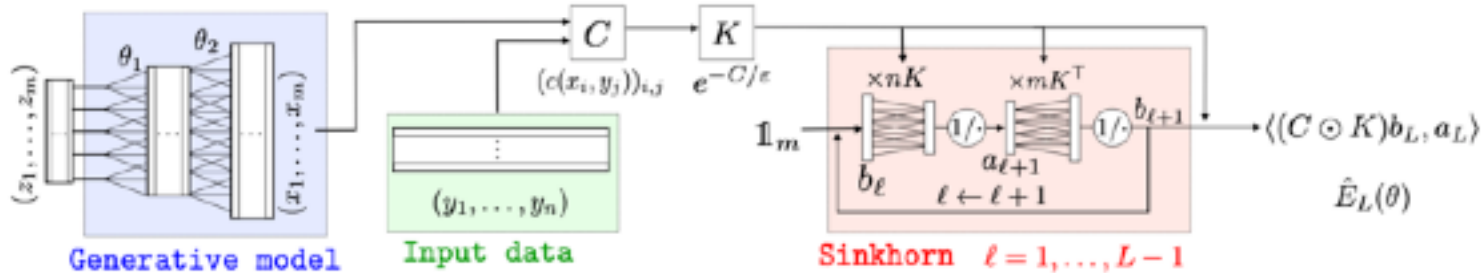
Université Paris-Saclay

Abstract

The ability to compare two degenerate probability distributions, that is two distributions supported on low-dimensional manifolds in much higher-dimensional spaces, is a crucial

1 Introduction

Several important statistical problems boil down to fitting densities, *i.e.* estimating the parameters of a chosen model that *fits* observed data in some meaningful way. While the standard approach is maximum likelihood estimation, this approach is often flawed in



Sinkhorn AutoEncoders

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Tim Genewein‡

Bosch Center for
Artificial Intelligence

Frank Nielsen

École Polytechnique
Sony CSL

Applications in ML

- Most medium-to-large scale applications of OT in ML!
- Generative Modeling
 - **Sinkhorn GAN**: uses entropic OT as a training loss → faster, stable optimization.
 - Provides a differentiable surrogate to Wasserstein distance.
- Representation Learning
 - Align embeddings across languages or modalities with regularized OT.
 - Benefit: couplings spread mass → avoid overfitting to noise.

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The ability to compare two degenerate probability distributions, that is two distributions supported on low-dimensional manifolds in much higher-dimensional spaces, is a crucial

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Generative model Input data Sinkhorn $\ell = 1, \dots, L-1$

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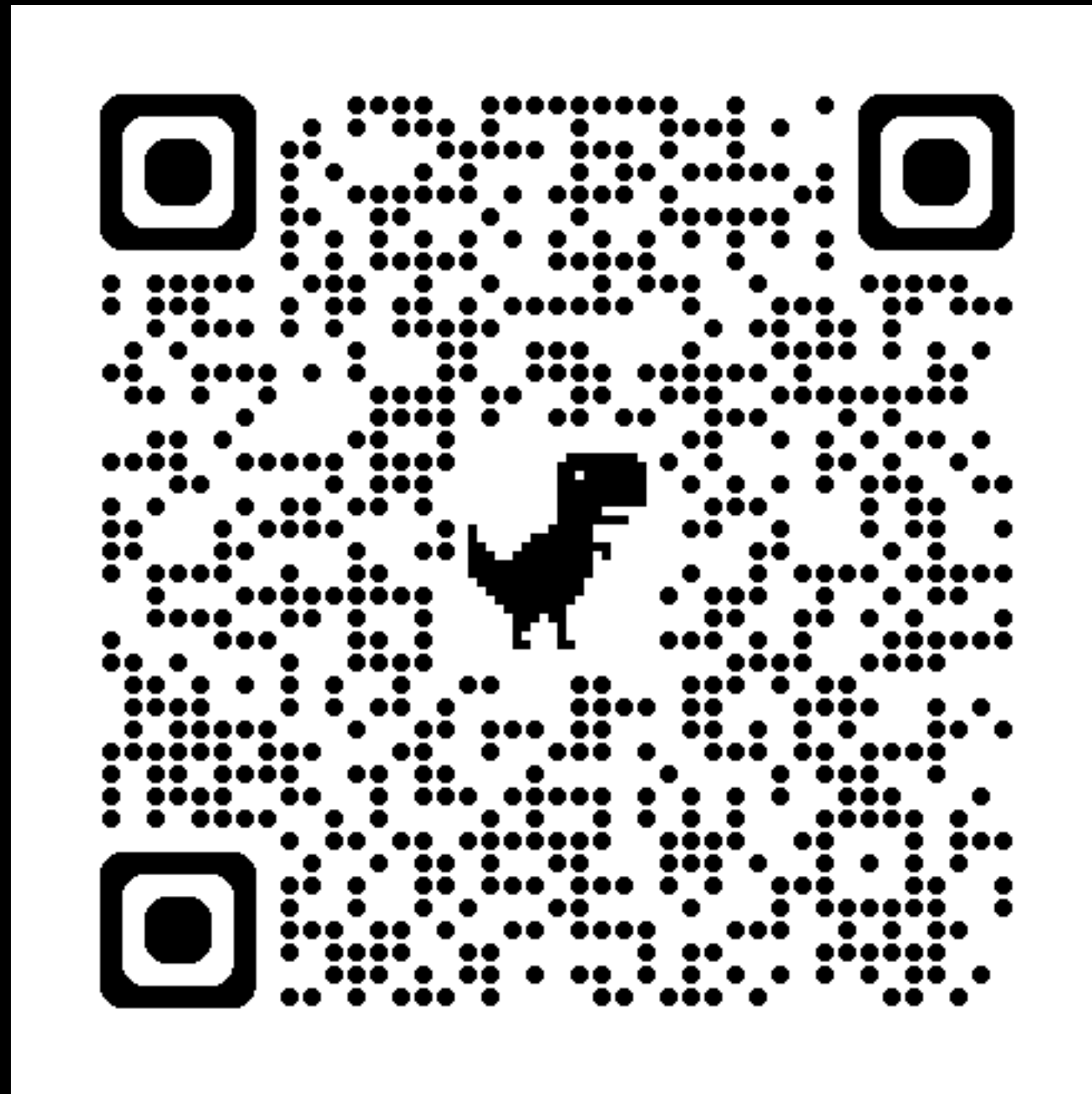
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Interactive Demo: Sinkhorn

1. Implement pairwise distance function
2. Implement Sinkhorn's algorithm
3. Verify feasibility of solution + visualize



tinyurl.com/cs2840-lecture5

Sinkhorn Algorithm

1. $\mathbf{K} \leftarrow \exp\{-\mathbf{C}/\varepsilon\}$
2. Repeat until convergence:
 $\mathbf{u} \leftarrow \mathbf{a} \oslash \mathbf{K}\mathbf{v}, \mathbf{v} \leftarrow \mathbf{b} \oslash \mathbf{K}^T\mathbf{u}$

Tips: to avoid numerical issues you can:

- Add small values before dividing
- [Even better] compute in log domain